

CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY

Fourth Semester of B. Tech. Examination (CL)

December 2011

CL 210 Hydrology and Hydraulics

Date: 08.12.2011, Thursday

Time: 10:00 a.m. To 01:00 p.m.

Maximum Marks: 70

Instructions:

1. The question paper comprises of two sections.
2. Section I and II must be attempted in separate answer sheets.
3. Make suitable assumptions and draw neat figures wherever required.
4. Use of scientific calculator is allowed.

SECTION - I

- Q - 1 (a) Distinguish between 'Recording' and 'Non-recording' type of rain-gauges. [02]
- (b) The isohyets interval and area with respect to isohyets due to a storm in catchment are mentioned in table given below. The total drainage basin area is of 600 km². Estimate the mean precipitation due to storm. [05]

Isohyetal interval (cm)	15-12	12-9	9-6	6-3	3-1
Inter-isohyetal area (km ²)	92	128	120	175	85

- Q - 2 (a) Determine the annual runoff and annual runoff coefficient for a catchment by Khosla's formula. The mean monthly rainfall (P_m) and Temperature (T_m) are given below. The annual rainfall in area is 104 cm. [06]

Month	Jan.	Feb.	Mar.	Apr.	May	June	July	Aug.	Sep.	Oct.	Nov.	Dec.
T_m (°C)	10	15	20	25	30	35	32	30	25	28	15	12
P_m (cm)	3	3	2	0	2	10	35	28	15	3	1	2

- (b) Define: Transpiration and Evapotranspiration. What are the factors affecting evapotranspiration? [05]
- (c) Explain Pan coefficient. Enlist various methods of reducing evaporation from a reservoir? [03]

OR

- Q - 2 (a) Find out the ordinates of DRH due to rainfall excess of 4.5 cm occurring in 6 hr. The ordinates of a 6-hr unit hydrograph for a catchment are given in table. [06]

Time (hrs)	0	3	6	9	12	15	18	21	24	30	36	42	48
UH Ordinate (cumecs)	0	70	100	150	200	140	130	120	100	80	42	20	0

Show the ordinates of calculated DRH on graph paper.

- (b) Define Infiltration. Explain Infiltration process briefly. What are the factors affecting infiltration capacity? [05]
- (c) Write a short note on double ring infiltrometer. [03]

- Q - 3 (a) A 30 cm well fully penetrates into a confined aquifer 30 m deep. After a long period of pumping at a rate of 1300 lpm, the drawdown in the wells at 22 m and 44 m from the pumping well are found to be 2.0 and 1.5 m, respectively. Calculate transmissibility and drawdown in the pumped well. [07]
- (b) Explain the following terms : [03]
- (i) Perched aquifer (ii) Artesian well (iii) Coefficient of permeability
- (c) What are the methods for estimation of yield of an open well? Explain any one of them in detail. [04]

OR

- Q - 3 (a) A 25 cm well fully penetrates 30 m below the static water table. After a long period of pumping at a rate of 1500 lpm, the drawdown in the wells at 20 m and 40 m from the pumped well were found to be 3.5 m and 0.9 m, respectively. Determine the transmissibility of the aquifer. [06]
- (b) Derive an expression for the discharge from a well in an unconfined aquifer based on Dupit's theory. [04]
- (c) Write short notes on following ground water investigation methods: [04]
- (i) Electrical resistivity (ii) Radio active logging

SECTION - II

- Q - 4 What is salt water intrusion? How it is caused? Explain the concept of slope and shape of interface. [7]
- Q - 5 (a) What are the needs of ground water recharge? [2]
- (b) Discuss physical qualities of ground water briefly. [5]
- (c) What is ground water pollution? Discuss the pollution mechanism briefly. Which are the factors affecting ground water pollution? [7]

OR

- Q - 5 (a) Classify the methods of artificial recharge. [4]
- (b) Discuss briefly chemical qualities of ground water. [3]
- (c) Discuss the sources and nature of pollution from Industrial and Agricultural wastes. [4]
- (d) Write a short note on pollution from wells and salt water. [3]
- Q - 6 (a) Define: Aquifer, Aquiclude, Aquifuge, Aquitard, Transmissibility and Storage Coefficient [6]
- (b) What is Conjunctive Use? Why it is needed? [5]
- (c) What are the advantages and disadvantages of roof rain water harvesting system? [3]

OR

- Q - 6 (a) Explain constant head and falling head permeability measurement methods in detail. [6]
- (b) Estimate the storage coefficient of the aquifer for data given below: [5]
- $b = 36 \text{ m}$, $\eta = 12\%$, $\alpha = 10^{-7} \text{ N/m}^2$, $K_w = 2.3 \times 10^9 \text{ N/m}^2$ and $\gamma_w = 9810 \text{ N/m}^3$.
- (c) Explain the 'water conservation'. Discuss in detail the techniques used to conserve the surface runoff. [3]
